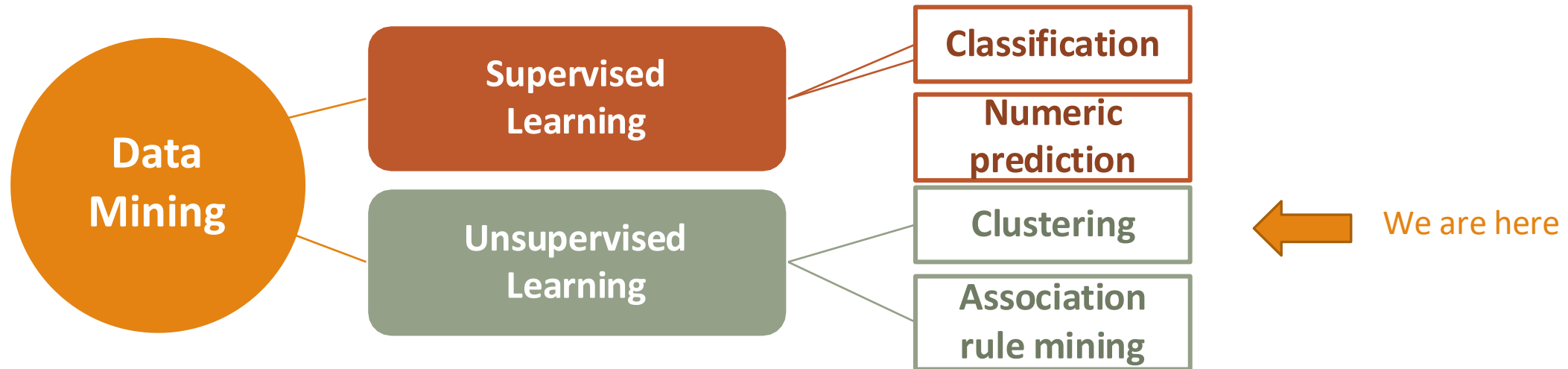


Lecture 9: Unsupervised Learning: Clustering

Outline

- Unsupervised learning
- Clustering
 - Understanding Clustering
 - The k-means clustering algorithm
 - Elbow Method
 - Classification with Clustering

Data Mining Tasks

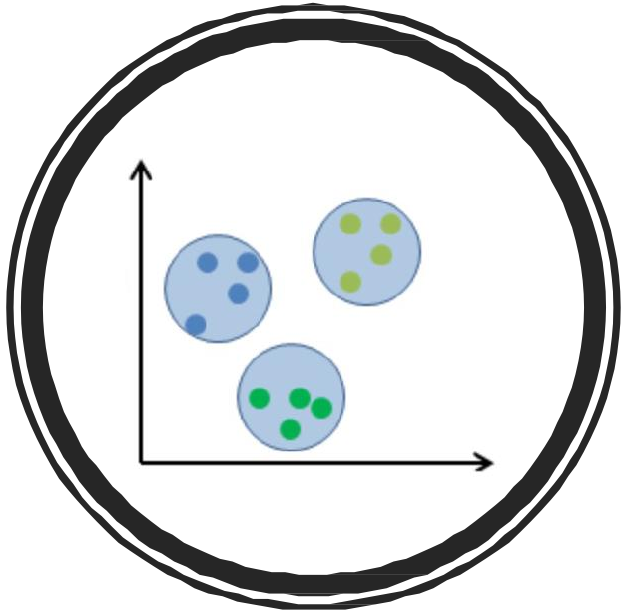


Data Mining Tasks

- **Supervised learning** is where you have input variables (x) and an output variable (Y) and you use an algorithm to learn the mapping function from the input to the output.
 - $Y=f(X); X \xrightarrow{\text{predict}} Y$
 - A **predictive model** is used for tasks that involve, as the name implies, the prediction of one value using other values in the dataset.
 - Model training -> model evaluation

Data Mining Tasks

- **Unsupervised learning** is where you only have input data (X) and no corresponding output variables.
- A **descriptive model** is used for tasks that would benefit from the insight gained from **summarizing data in new and interesting ways**.
- There is no target to learn, the process of training a descriptive model is called unsupervised learning.



Unsupervised Learning - Clustering Analysis

First, Let's Talk about American Express



Credit card giant American Express offers different types of cards:

- High annual fee, earn points on booking flights and hotels.
- No annual fee, cash back in supermarkets and restaurants.

First, Let's Talk about American Express



Credit card giant American Express offers different types of cards:

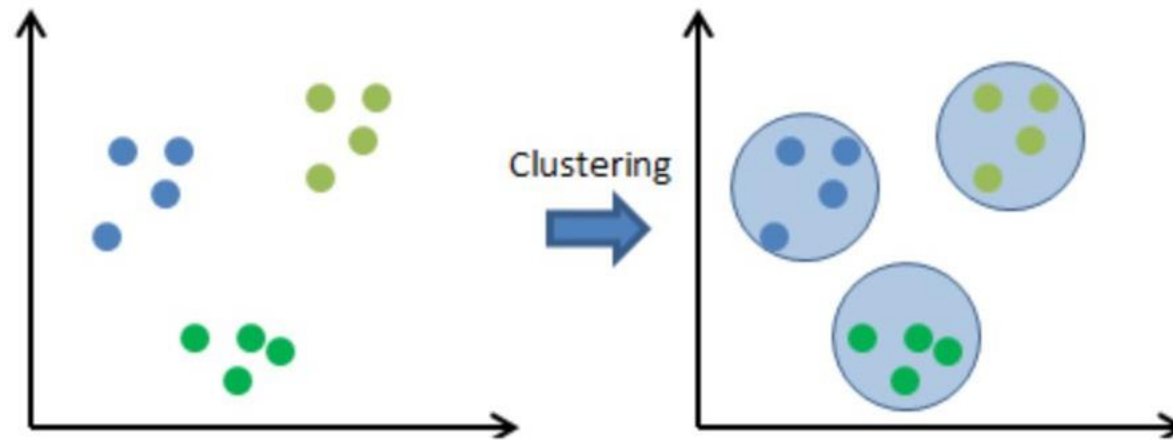
- High annual fee, earn points on booking flights and hotels.
- No annual fee, cash back in supermarkets and restaurants.



- **Customer Segmentation**

What is clustering?

- Clustering analysis: A process to segment a group of objects (examples) into multiple distinct **clusters**, such that all the members in one cluster are **similar** to each other and **distinctively different** from the members of any other clusters.



How Can Clustering Benefit Business?



Business Applications

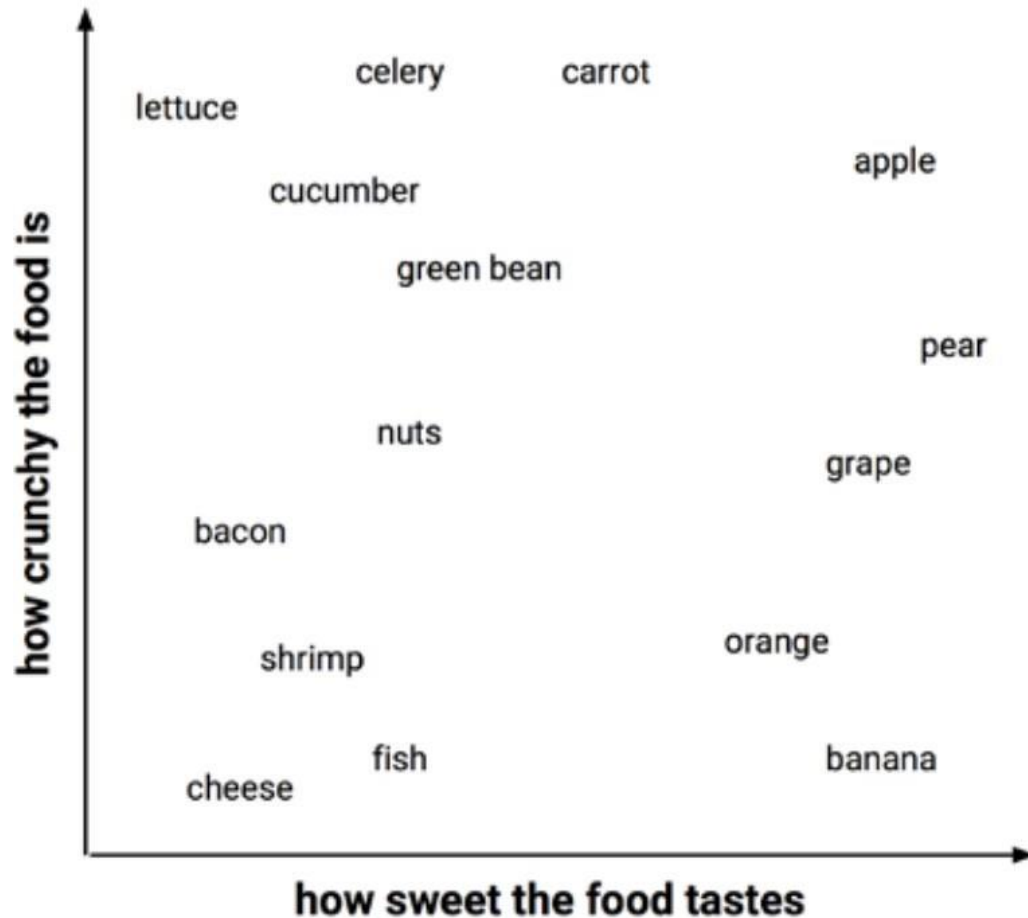
- 1. Customer Segmentation:** segmenting customers into groups with similar demographics or buying patterns for targeted marketing campaigns.
- 2. Industry analysis:** finding groups of similar firms based on measures such as growth rate, profitability, market size, and product range, so that we can better understand industry structure and to determine, for instance, who is a competitor.
- 3. Creating balanced portfolios:** Given data on a variety of investment opportunities (e.g., stocks), one may find clusters based on financial performance variables such as return (daily, weekly, or monthly), volatility, beta, and market capitalization. Selecting securities from different clusters can help create a balanced portfolio.

The k-means Clustering Algorithm

- The k-means algorithm
 - k is the number of clusters
 - Assigns each of the examples to one of the k clusters (**clusters are not overlapped**)
 - The goal is to minimize the differences within each cluster and maximize the differences between the clusters

The k-means Clustering Algorithm

- Food ingredients described by sweetness and crunchiness



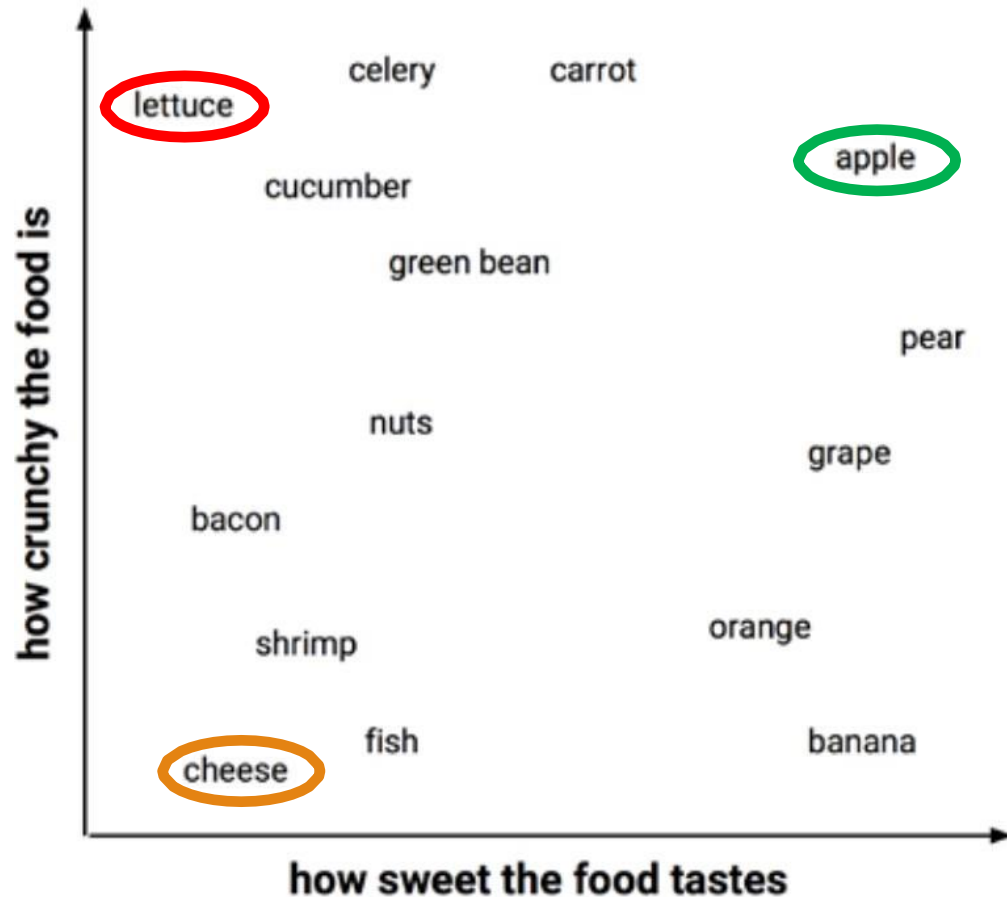
Two features

- Sweetness: 1—10
- Crunchiness : 1—10

There are 3 food types

The k-means Clustering Algorithm

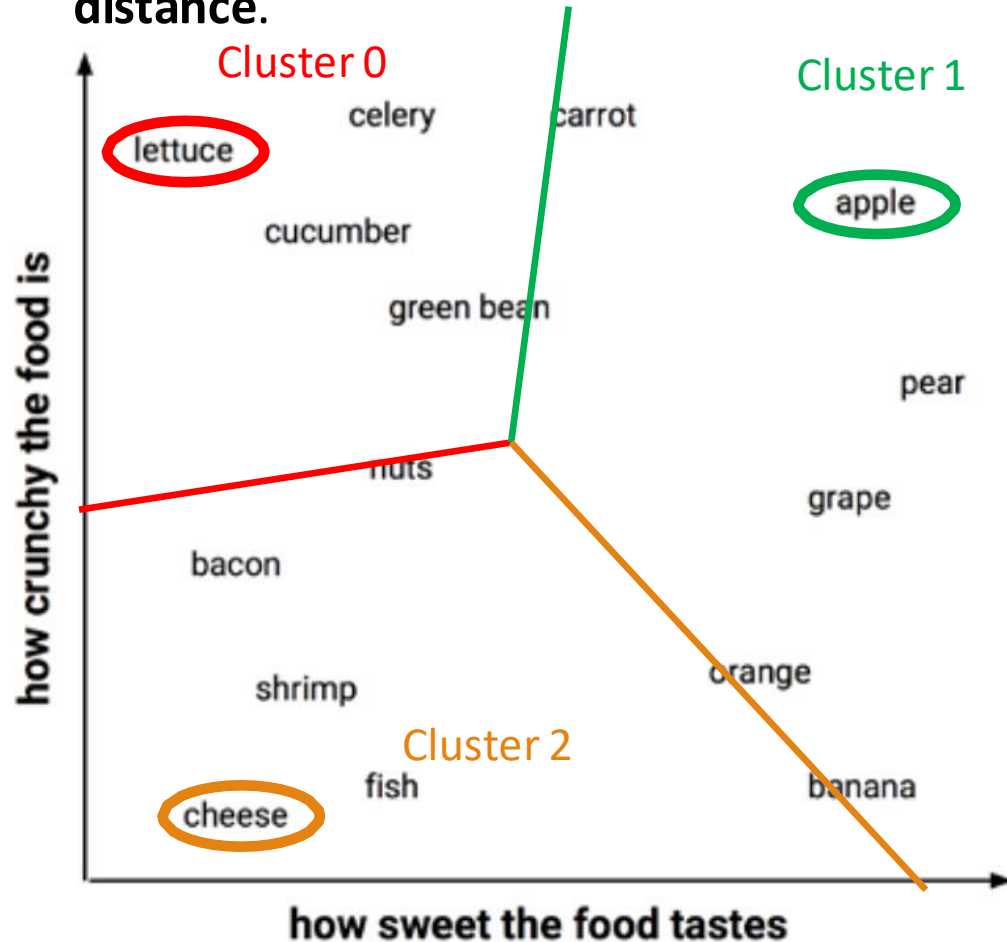
- Step 1: Randomly choosing k points in the feature space to serve as the cluster centers.



$k = 3$: 3 data points are randomly selected

The k-means Clustering Algorithm

- Step 2: Assigning the remaining examples to the cluster center that is nearest according to the **distance**.



The k-means Clustering Algorithm

- Step 2: Assigning the remaining examples to the cluster center that is nearest according to the distance.

Euclidean distance between example x and example y

$$\text{dist}(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

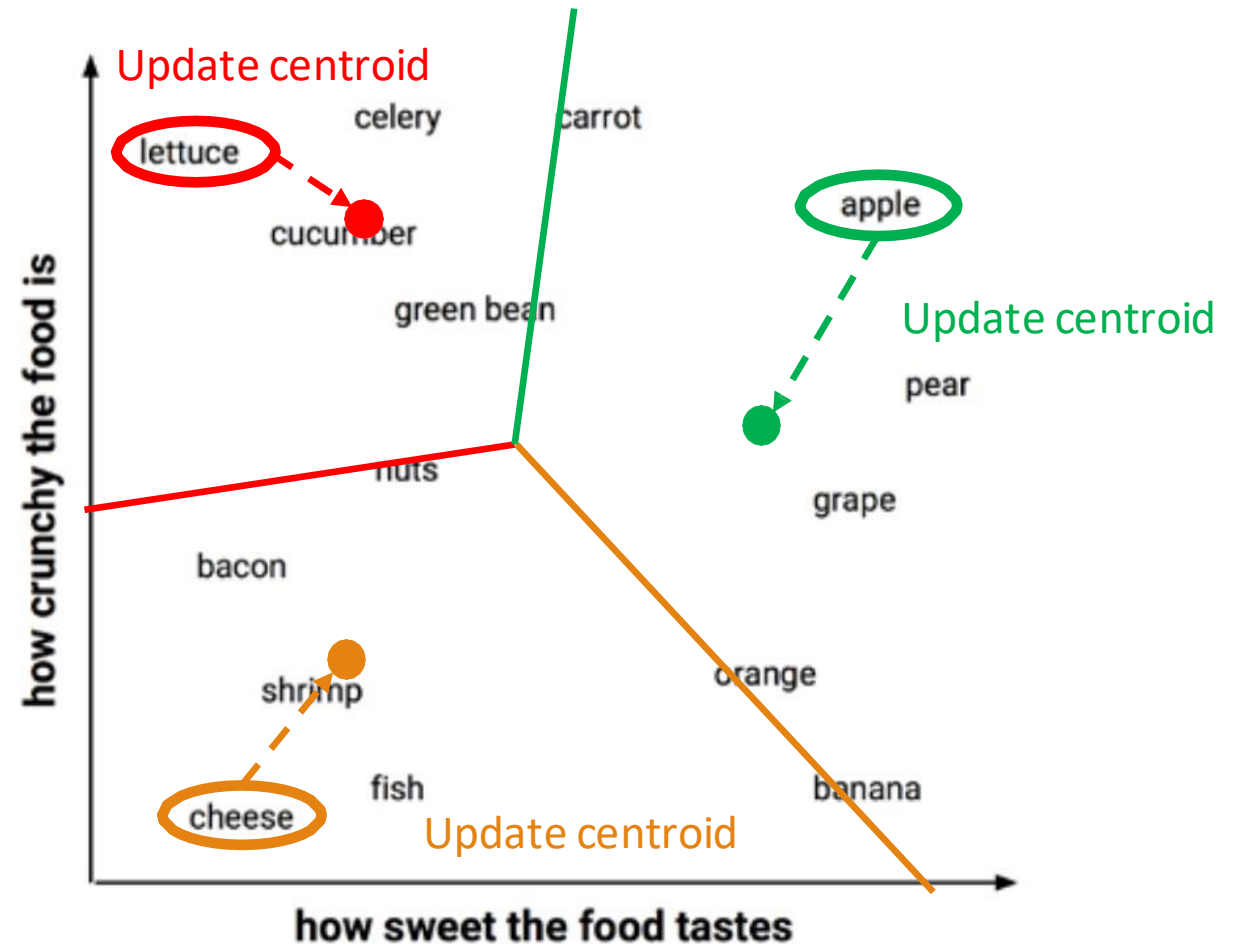
ingredient	sweetness	crunchiness
grape	8	5
green bean	3	7

$$\text{dist}(\text{grape}, \text{green bean}) = \sqrt{(8-3)^2 + (5-7)^2} = 5.4$$

Using this function, we find the distance between each example and each cluster center. The example is then assigned to the nearest cluster center.

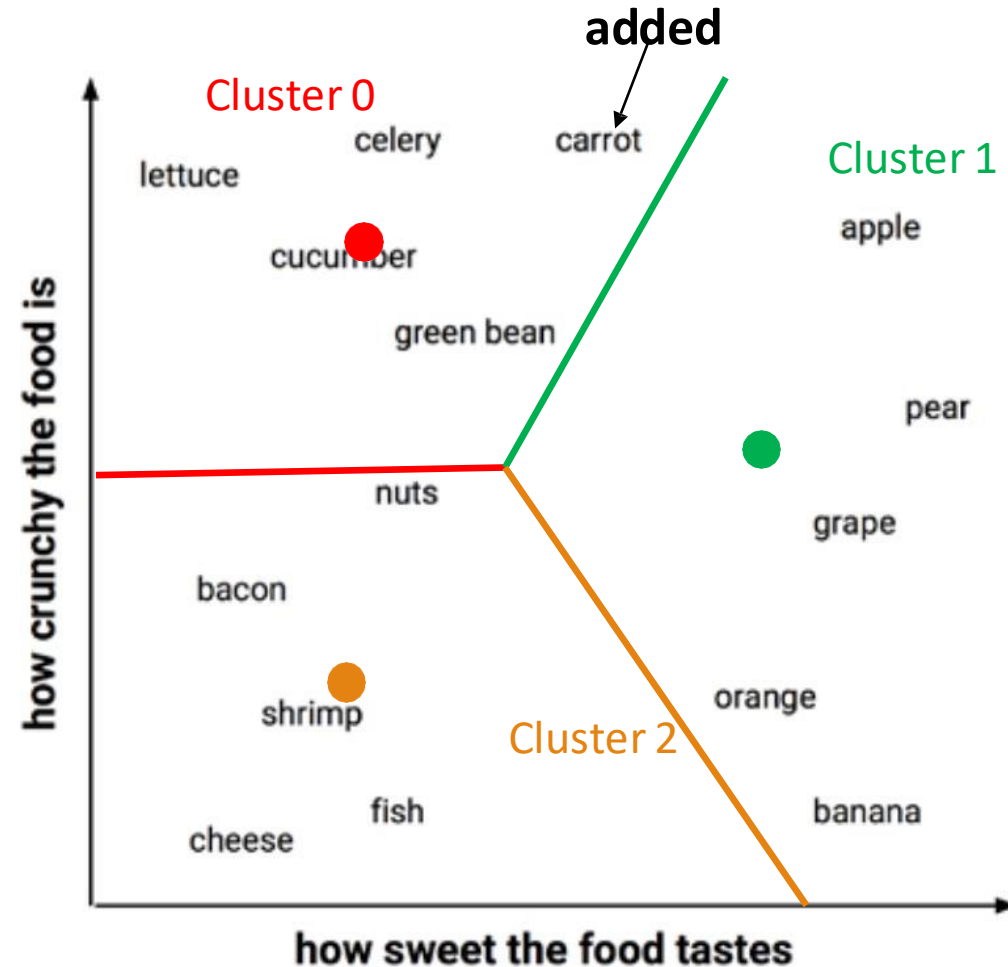
The k-means Clustering Algorithm

- Step 3: Shifting the initial centers to a new location, known as the **centroid**, which is calculated as the average position of the points currently assigned to that cluster.



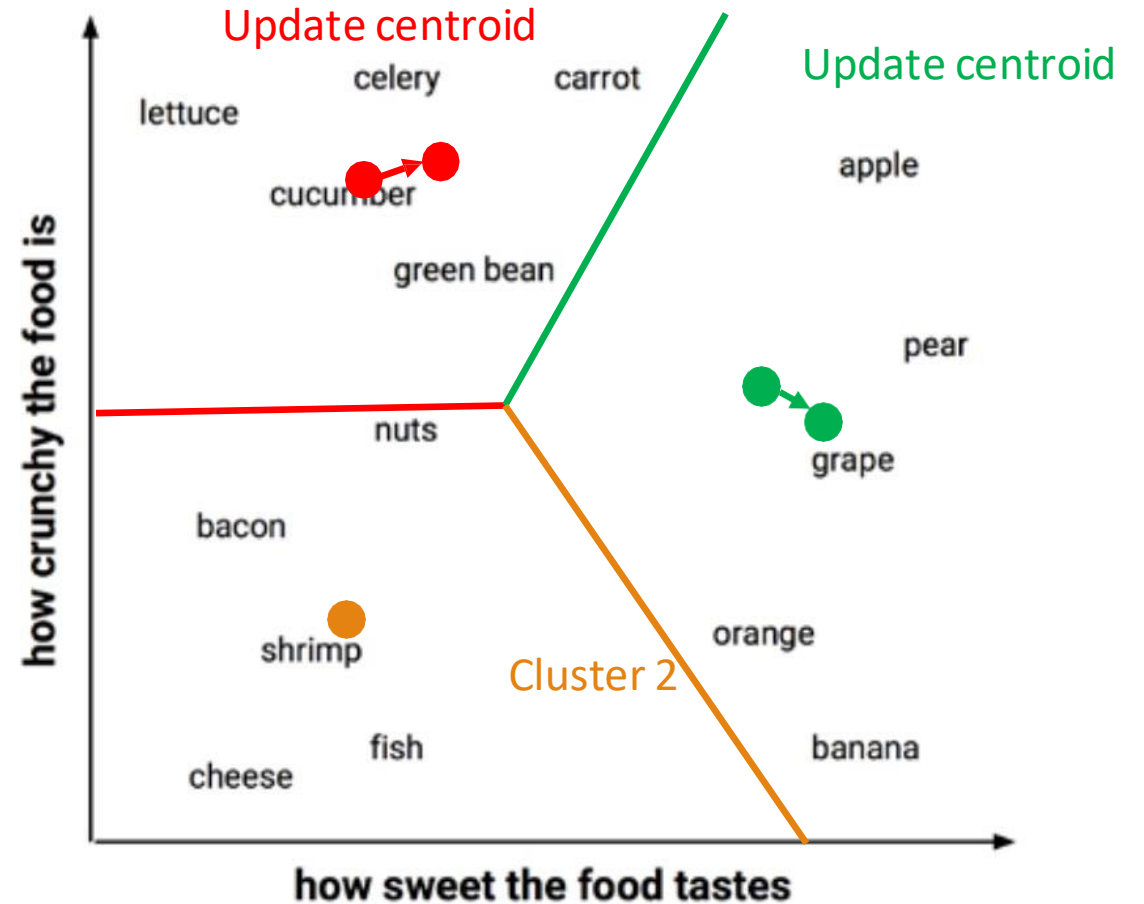
The k-means Clustering Algorithm

- Repeat step 2: Assigning the remaining examples to the cluster center that is nearest according to the distance.



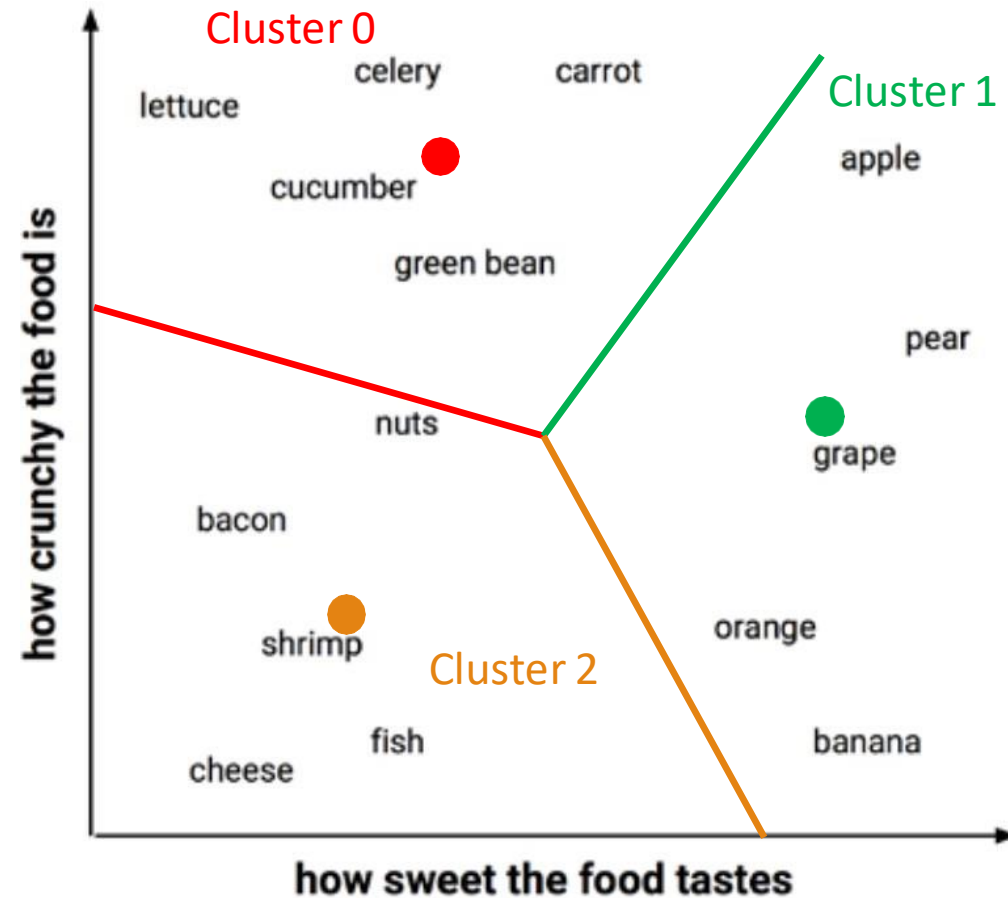
The k-means Clustering Algorithm

- Repeat step 3: Shifting the centroids to a new locations.



The k-means Clustering Algorithm

- Repeat step 2: Assigning the remaining examples to the cluster center that is nearest according to the distance.



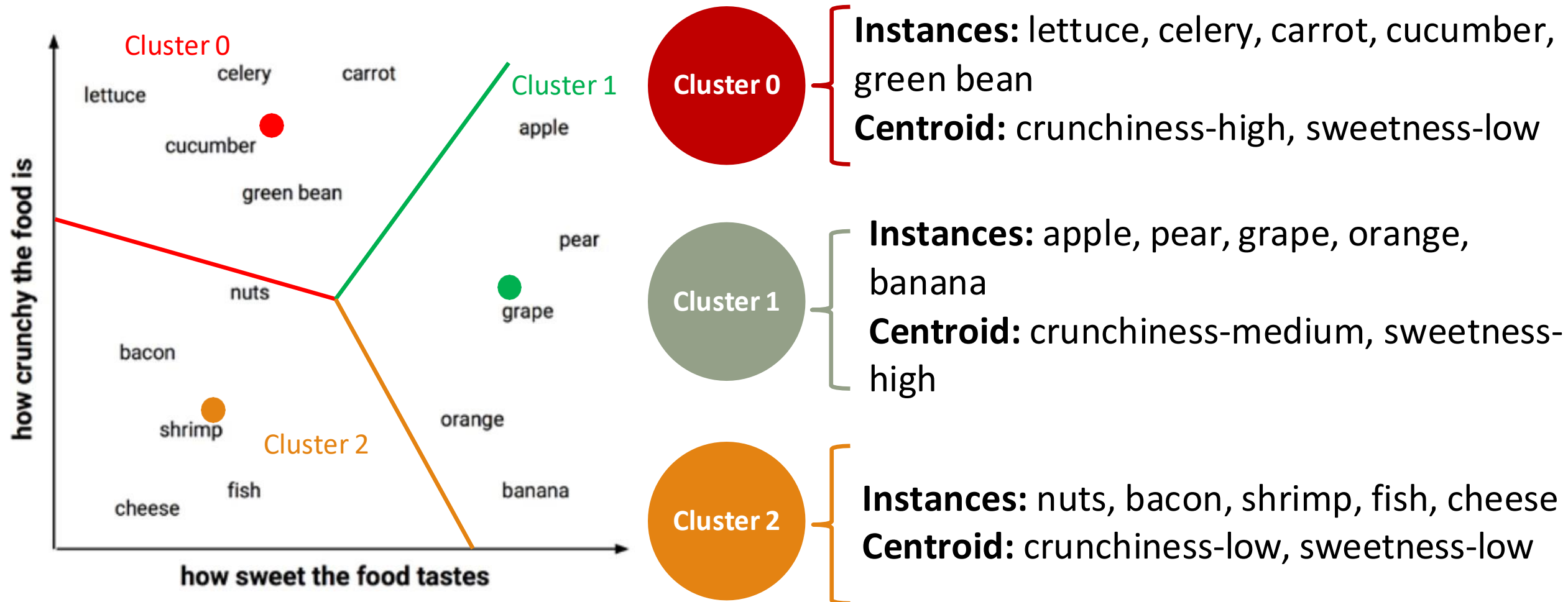
The k-means Clustering Algorithm

- Step 1: Randomly select k points in the feature space to serve as the cluster centers.
- Step 2: Assigning the remaining examples to the cluster that is nearest according to the distance.
- Step 3: Recalculate the centroid of each cluster
- Step 4: Repeat step 2 and step 3 until the centroids stop shifting.

The k-means Clustering Algorithm

- Clustering results
 - The cluster assignments of each example (**size of cluster**)
 - Cluster centroids (**cluster center**)
- There is no training and testing process for unsupervised learning.

The k-means Clustering Algorithm



The k-means Clustering Algorithm (Disadvantage)

- k-means algorithm is highly sensitive to the starting position of the cluster centers.
 - Random chance may have a substantial impact on the final set of clusters.
- k-means++ proposes an alternative method for selecting the initial cluster centers (the default setting in SK-learn).

Probability of selecting point x_i

$$P(x_i) = \frac{D(x_i)^2}{\sum_{j=1}^n D(x_j)^2}$$

Distance from point x_i to its
nearest already chosen center



Clustering on BART Riders

- You have been given a data file by the San Francisco Bay Area Rapid Transit (BART), which identifies a set of demographics for residents in a local area.
- We will use this file to determine residents' segmentations so that we can use it to develop marketing plans accordingly.

Variable	Description
Age:	1. 14 thru 17; 2. 18 thru 24; 3. 25 thru 34; 4. 35 thru 44; 5. 45 thru 54; 6. 55 thru 64; 7. 65 and Over
DistToWork:	Distance to work in miles
DualInc:	Is dual income household or not
Education:	1. Grade 8 or less; 2. Grades 9 to 11; 3. Graduated high school; 4. 1 to 3 years of college; 5. College graduate; 6. Grad Study
Gender:	M or F
Income:	1. Less than \$10,000; 2. \$10,000 to \$14,999; 3. \$15,000 to \$19,999; 4. \$20,000 to \$24,999; 5. \$25,000 to \$29,999; 6. \$30,000 to \$39,999; 7. \$40,000 to \$49,999; 8. \$50,000 to \$74,999; 9. \$75,000 or more
Language:	Language spoken at home, English, Spanish, Other
NbrInHouseHold:	Number in household
NbrInHouseholdUnder18:	Number in household under 18 years old
OwnRent:	Own, rent, or live with parents
YrsInArea:	Years lived in bay area
Rider:	No, Non-rider; Yes, Rider

Clustering on BART Riders

	Age	DistToWork	DualInc	Education	Gender	Income	Language	NbrInHouseHold	NbrInHouseholdUnder18	OwnRent	YrsInArea	Rider
0	7	14	N	3	F	3	English	1	0	Rent	5	Yes
1	7	10	N	5	M	8	English	2	0	Own	5	No
2	3	9	N	3	M	1	English	1	0	Rent	5	Yes
3	1	13	N	2	M	1	English	5	3	Parent	5	Yes
4	3	14	N	5	F	2	English	3	1	Parent	5	Yes
...	...	...	...	...	...	...	...	...	...	...	...	...
5488	3	11	Y	4	M	5	English	2	0	Own	5	Yes
5489	4	9	Y	3	F	6	English	3	1	Rent	2	Yes
5490	2	7	N	4	M	2	English	1	1	Rent	1	Yes
5491	3	7	Y	3	F	9	English	2	0	Own	5	No
5492	2	10	N	4	F	1	English	4	2	Parent	2	Yes

5493 rows x 12 columns

Clustering on BART Riders

- Create dummy variables
- Apply minmax normalization

Age	DistToWork	Education	Income	NbrInHouseHold	NbrInHouseholdUnder18	YrsInArea	DualInc_Y	Gender_M	Language_Other	Language_Spanish	OwnRent_Parent	OwnRent_Rent	Rider_Yes
1.000000	0.647059	0.4	0.250	0.000	0.000000	1.00	0.0	0.0	0.0	0.0	0.0	1.0	1.0
1.000000	0.411765	0.8	0.875	0.125	0.000000	1.00	0.0	1.0	0.0	0.0	0.0	0.0	0.0
0.333333	0.352941	0.4	0.000	0.000	0.000000	1.00	0.0	1.0	0.0	0.0	0.0	1.0	1.0
0.000000	0.588235	0.2	0.000	0.500	0.333333	1.00	0.0	1.0	0.0	0.0	1.0	0.0	1.0
0.333333	0.647059	0.8	0.125	0.250	0.111111	1.00	0.0	0.0	0.0	0.0	1.0	0.0	1.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...
0.333333	0.470588	0.6	0.500	0.125	0.000000	1.00	1.0	1.0	0.0	0.0	0.0	0.0	1.0
0.500000	0.352941	0.4	0.625	0.250	0.111111	0.25	1.0	0.0	0.0	0.0	0.0	1.0	1.0
0.166667	0.235294	0.6	0.125	0.000	0.111111	0.00	0.0	1.0	0.0	0.0	0.0	1.0	1.0
0.333333	0.235294	0.4	1.000	0.125	0.000000	1.00	1.0	0.0	0.0	0.0	0.0	0.0	0.0
0.166667	0.411765	0.6	0.000	0.375	0.222222	0.25	0.0	0.0	0.0	0.0	1.0	0.0	1.0

Clustering on BART Riders

- KMeans(n_clusters=2, random_state=0)

Counter({0: 3159, 1: 2334})

Cluster size

	cluster 1	cluster 2
Age	4.163976	2.562982
DistToWork	11.533713	11.413882
Education	4.361823	3.209940
Income	6.870529	2.846187
NbrInHouseHold	2.646724	3.255356
NbrInHouseholdUnder18	0.566635	0.897601
YrsInArea	4.330801	4.239931
DualInc_Y	0.377018	0.063839
Gender_M	0.475783	0.442159
Language_Other	0.014562	0.050557
Language_Spanish	0.027857	0.092545
OwnRent_Parent	0.033238	0.556127
OwnRent_Rent	0.338398	0.269923
Rider_Yes	0.023742	0.976435

Centroids

Clustering on BART Riders

- KMeans(n_clusters=3, random_state=0)

Counter({0: 1486, 1: 1688, 2: 2319})

Cluster size

	cluster 1	cluster 2	cluster 3
Age	1.875505	3.234005	4.695990
DistToWork	11.423957	11.498815	11.508840
Education	2.765141	4.052725	4.450625
Income	2.881561	4.405213	7.170763
NbrInHouseHold	3.936070	2.164100	2.784390
NbrInHouseholdUnder18	1.246972	0.338270	0.630013
YrsInArea	4.429341	3.777844	4.578698
DualInc_Y	0.002692	0.152844	0.464856
Gender_M	0.421938	0.563389	0.412678
Language_Other	0.055182	0.026659	0.015955
Language_Spanish	0.098923	0.051540	0.030185
OwnRent_Parent	0.921938	0.000000	0.014230
OwnRent_Rent	0.000000	0.997630	0.006468
Rider_Yes	0.914536	0.385664	0.148340

Centroids

Clustering on BART Riders

- KMeans(n_clusters=3, random_state=0)

Counter({0: 1486, 1: 1688, 2: 2319})

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Centroids

Clustering on BART Riders

```
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```

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Young Professionals
Cluster 1



Students
Cluster 0

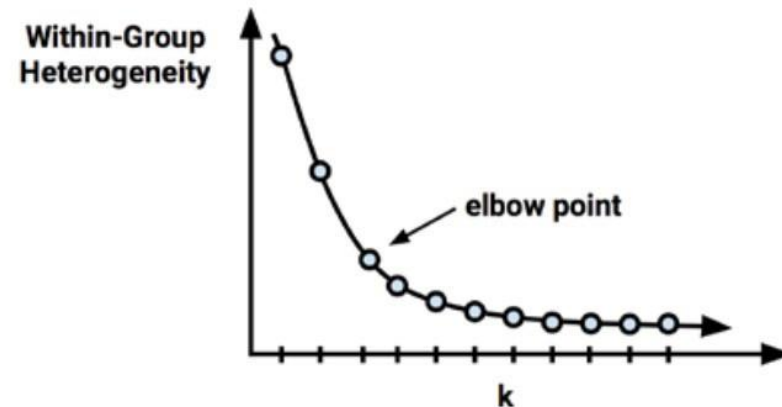
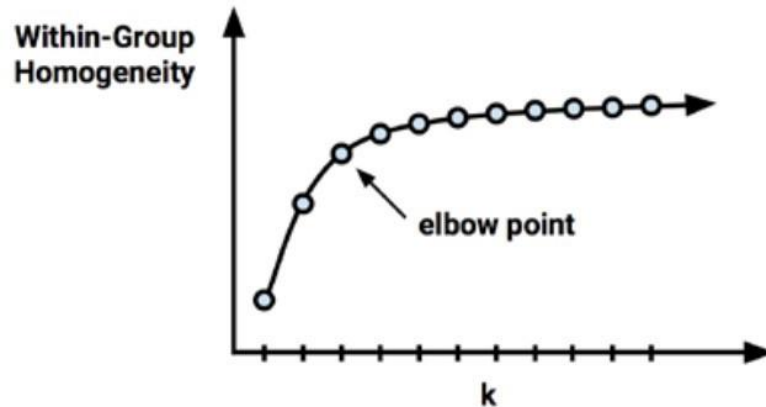


Managers
Cluster 2

If we segment residents into 3 clusters, what marketing plans you can use to target each cluster?

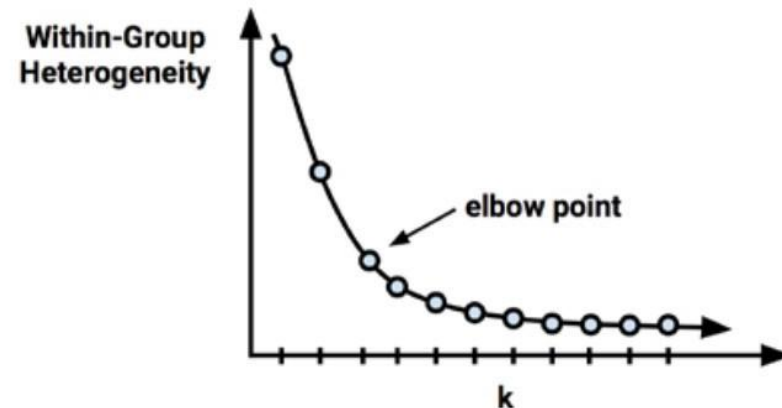
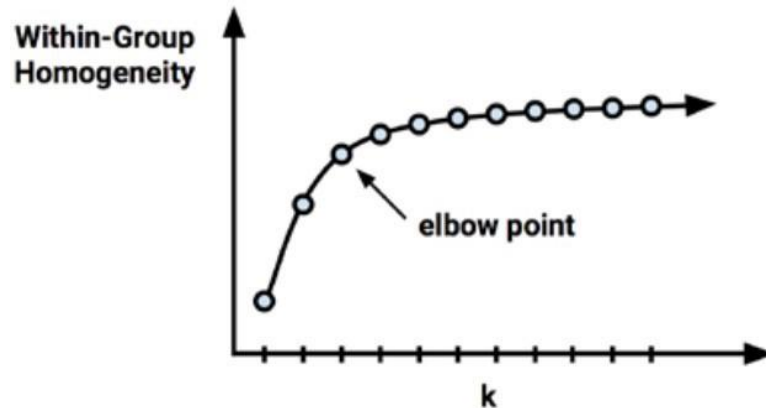
Elbow Method

- Choosing the appropriate number of clusters: **elbow method**
 - **Elbow method** attempts to gauge how the homogeneity or heterogeneity within the clusters changes for various values of k .
 - The homogeneity within clusters is expected to increase as additional clusters are added; heterogeneity will continue to decrease with more clusters.



Elbow Method

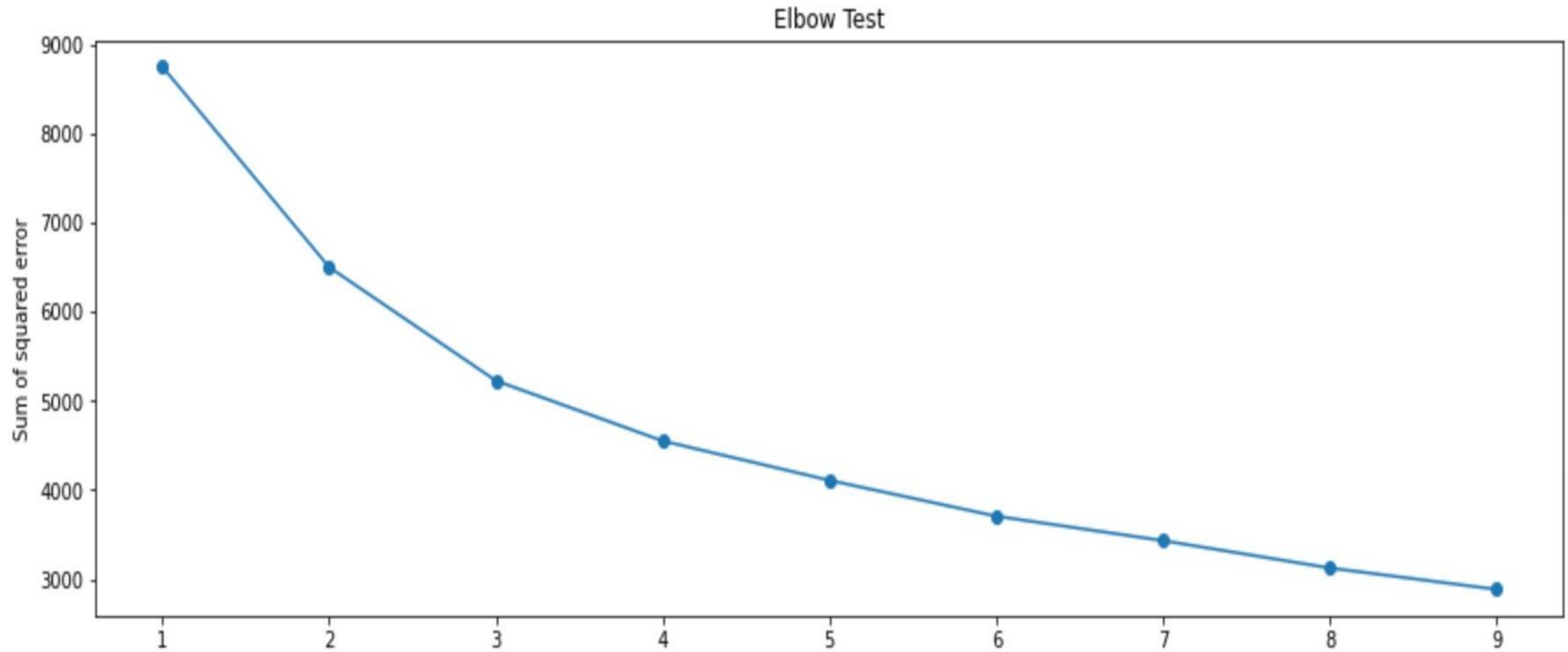
- Choosing the appropriate number of clusters: **elbow method**
 - You could continue to see improvements until each example is in its own cluster, the goal is not to maximize homogeneity or minimize heterogeneity, but rather to find k so that there are diminishing returns beyond that point. This value of k is known as the **elbow point** because it looks like an elbow .



Elbow Method

- Choosing the appropriate number of clusters: **elbow method**
 - Measure homogeneity or heterogeneity within the clusters
 - Within-cluster sum of squared errors: the squared errors from the mean (centroid) of the cluster of all the observations belonging to that cluster. The Within-cluster sum of squared errors $W(C_k)$ of a cluster C_k is defined as $\sum_{x_i \in C_k} (x_i - \bar{x}_k)^2$, where $\bar{x}_k$ is the mean (centroid) of cluster C_k .

Elbow Method



Silhouette Score (Clustering Evaluation)

- Measures how well each data point fits within its assigned cluster
- Compares:
 - Cohesion** → how close the point is to its own cluster
Average distance from the point to all other points in its own cluster (a)
 - Separation** → how far it is from other clusters
Average distance from the point to points in the closest other cluster (b)

Score range:

- +1 → well-separated, correctly clustered
- 0 → overlapping clusters
- -1 → likely assigned to the wrong cluster

$$\text{silhouette} = \frac{b - a}{\max(a, b)}$$

Interpretation: Higher average silhouette score = better-defined and more distinct clusters

Classification vs. Clustering

- Classification is supervised with labeled data
- KNN vs K-Means
- Classification with Clustering
 - Step 1: clustering
 - Step 2: build a classification model for each cluster

Clustering vs Topic Modeling

Topic Modeling is an unsupervised machine learning technique used to identify hidden themes (topics) in a **collection of text documents** by discovering patterns of word co-occurrence.

- Both aim to discover hidden structure in data (unsupervised learning)
- Both group data based on similarity patterns
- Topic modeling can be seen as **soft clustering** for **text**
- Clustering → one group per item
- Topic modeling → multiple topics per document (with probabilities)
- How to use them together:
 - Use topic modeling to extract features (topic proportions)
 - Then apply clustering to group similar documents or users
 - Use profile analysis to interpret results